

Quiz (Carolina Reaper)

- Q1.** What is the vertex of the parabola $y = x^2 + 4x + 4$?
(A) (0, 0)
(B) (0, 4)
(C) (-2, 0)
(D) (-2, -4)
(E) (-4, 0)
- Q2.** Which of the following is the axis of symmetry for $y = x^2 - 6x + 9$?
(A) $x = -3$
(B) $x = 0$
(C) $x = 3$
(D) $x = 6$
(E) $x = 9$
- Q3.** What are the x -intercepts of $y = (x + 2)(x - 4)$?
(A) (-2, 0), (4, 0)
(B) (-4, 0), (2, 0)
(C) (-2, 0), (-4, 0)
(D) (2, 0), (4, 0)
(E) (0, 0), (0, 0)
- Q4.** For the quadratic $y = -x^2 + 2x + 3$, what is the maximum value?
(A) -3
(B) -2
(C) 1
(D) 3
(E) 5
- Q5.** What is the equation of the axis of symmetry for $y = 2x^2 - 4x - 1$?
(A) $x = -1$
(B) $x = 0$
(C) $x = 1$
(D) $x = 2$
(E) $x = -0.5$
- Q6.** The parabola $y = x^2 - 4x + 3$ opens:
(A) up
(B) down
(C) left
(D) right
(E) none
- Q7.** What are the y -intercepts of $y = x^2 + 2x + 1$?
(A) (-1, 0)
(B) (0, 1)
(C) (1, 0)
(D) (0, -1)
(E) (0, 2)
- Q8.** For $y = (x - 1)^2 - 4$, what is the minimum value?
(A) -3
(B) -2
(C) -1
(D) -4
(E) 0

Open Ended Questions (FRQ)

- Q9.** Graph the parabola $y = x^2 - 2x - 3$ and identify its vertex, axis of symmetry, and x -intercepts.

- Q10.** For the quadratic $y = -2x^2 + 4x + 1$, find the maximum value and the x -value at which it occurs.

Chef's Tips

- Use graph paper for questions 9 and 10.
- Clearly label all parts of the graph.
- Show all work and calculations for free response questions.